



Calculating theoretical probabilities from Venn diagrams (one event)

Task A: Theoretical probabilities from Venn diagrams

1) Use the Venn diagram to write the probabilities.

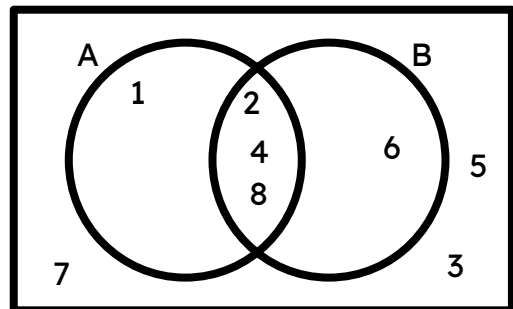
a) $P(A) =$

b) $P(B) =$

c) $P(\text{not } B) =$

d) $P(\text{only } A) =$

$$\xi = \{1, 2, 3, 4, 5, 6, 7, 8\}$$



2) Use the Venn diagram to write the probabilities.

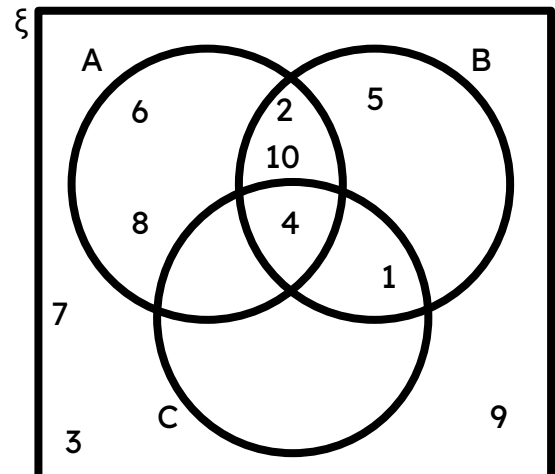
a) $P(A) =$

b) $P(B) =$

c) $P(C) =$

d) $P(\text{only } B) =$

e) $P(\text{not } C) =$



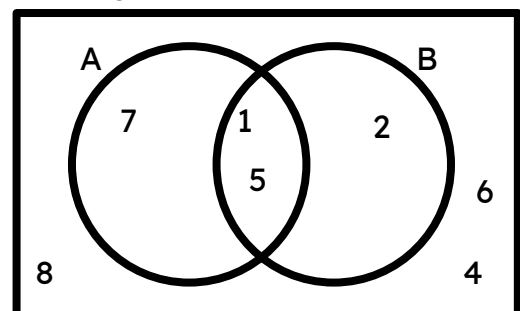
3)

a) Complete the Venn diagram given these probabilities.

$$P(A) = \frac{1}{2}$$

$$P(B) = \frac{3}{8}$$

$$\xi = \{\text{integers 1 to 8 inclusive}\}$$



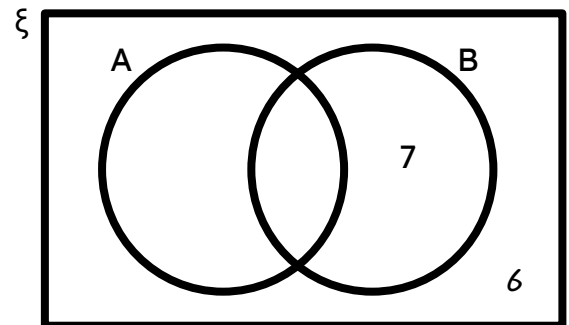
b) Work out $P(\text{only } A)$

4) This Venn diagram shows the number of outcomes in each event.

There are 21 outcomes in the sample space.

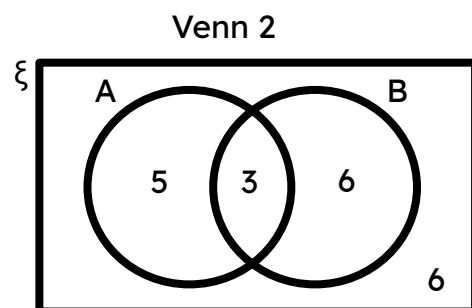
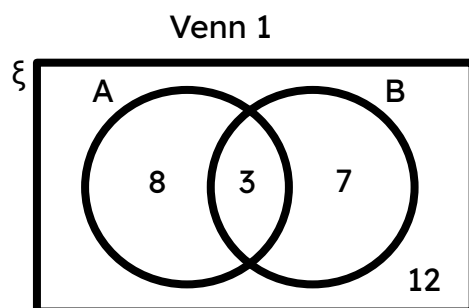
Complete the Venn diagram given these probabilities.

$$P(\text{not } A) = \frac{13}{21} \qquad P(\text{only } A) = \frac{1}{7}$$



Task B: Comparing probabilities

1) Compare the probabilities from each Venn diagram. The Venn diagrams show the number of outcomes.

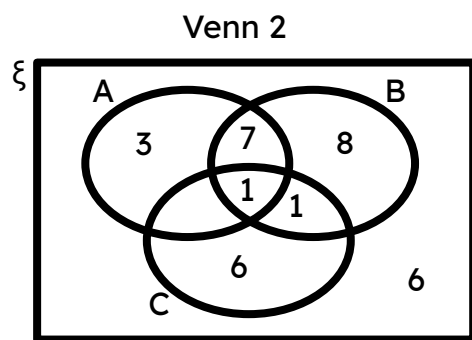
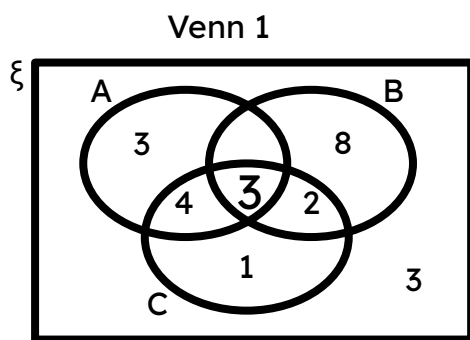


a) $P(A)$

b) $P(\text{only } A)$

c) $P(\text{not } B)$

2) Compare the probabilities from each Venn diagram. The Venn diagrams show the number of outcomes.



a) $P(A)$

b) $P(C)$

c) $P(\text{not } B)$

3) Complete the second Venn diagram with number of outcomes such that you are more likely to select an outcome from event B in Venn 2.

